



DepthWizard — Technology Stack Symbols

Monocular 2D Aerial/Satellite to 3D Digital Elevation Engine

SIH 2026 • PS-26175

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PART 1: ARTIFICIAL INTELLIGENCE, ML & COMPUTER VISION



Python

v3.10+

Core backend programming language powering tensor math, scientific routines, and async services.
Runtime: CPython 3.10+ • Type Annotations



PyTorch

v2.0+

Deep learning runtime powering forward inference for monocular relative depth estimation models.
torch, torch.hub, CUDA & CPU execution



OpenCV

v4.10

Computer vision processing: CLAHE enhancement, Canny edges, bilateral filtering & morphological segmentation.
opencv-python-headless • cv2 algorithms



Intel MiDaS / DPT

v2.1 / Small

Vision Transformer and convolutional relative depth estimator trained on 10+ diverse depth datasets.
Scale-and-shift invariant loss disparity



NumPy

v1.26.4

N-dimensional array engine for depth matrix manipulations, Cartesian 3D coordinates, and meshgrid.
Vectorized linear algebra • Float32 tensors



SciPy

v1.14.1

Scientific algorithms for spatial coordinate transformations, filtering, and connected-component calculations.
scipy.ndimage • scipy.spatial

PART 2: BACKEND FRAMEWORK, API & DATABASE



FastAPI

v0.115.0

High-performance async REST API framework powering all 11 DepthWizard endpoints, lifespan, and CORS.
AsyncIO • OpenAPI Docs • Starlette core



Uvicorn

v0.30.0

Lightning-fast ASGI web server implementation powering high-throughput asynchronous request loops.
uvloop & httptools event loop engine



SQLite /aiosqlite

v3.x / 0.20

Zero-configuration embedded relational SQL database with non-blocking async Python I/O driver.
depthwizard.db • ACID Transactional



SQLAlchemy

v2.0.35

Object-Relational Mapping (ORM) and declarative schema models for projects, jobs, and calibrations.
DeclarativeBase • AsyncSession queries



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PART 3: FRONTEND UI, 3D WEBGL & INTERACTIVE GRAPHICS



React

v18.3

Declarative component UI architecture powering SPA views, live telemetry, and reactive state updates.
Virtual DOM • React Suspense & Hooks



TypeScript

v5.5

Strictly typed frontend programming language preventing runtime errors across 3D vector coordinates.

Static Type System • Zero Type Errors



Three.js

r128+

Hardware-accelerated WebGL 3D graphics rendering engine for dense point clouds, meshes, and lighting.

THREE.BufferGeometry • Points & Mesh



React Three Fiber

v8.17

Declarative Three.js scene graph integration with React lifecycle, OrbitControls, and useFrame animation loops.

@react-three/fiber • @react-three/drei



Vite

v5.4

Next-generation frontend development server and Rollup production bundler with sub-second HMR.

Native ESM • Production bundle: 5.75s



Tailwind CSS

v3.4

Utility-first styling framework providing modern dark-mode aesthetics, responsive layouts, and animations.

JIT Compiler • PostCSS • Flexbox/Grid

PART 4: STATE MANAGEMENT, DEVOPS & REPORTING



Zustand

v4.5

Lightweight, boilerplate-free state management controlling active projects, 3D viewing modes, and camera timelines.

Zustand useStore • Unidirectional flow



Docker / Compose

v26.0+

Containerization specification enabling multi-container orchestration for FastAPI backend and React frontend.

Dockerfile • docker-compose.yml



Nginx

Alpine

High-concurrency reverse proxy, static file server, and API gateway routing between '/api/v1' and SPA.

Reverse Proxy • Gzip & Static caching



ReportLab

v4.2.2

Programmatic PDF document generator producing formal structural inspection reports and height audits.

PDF Export • Geospatial Summary